

Product Requirement Description

1. Product Overview

1.1 Vision

Build India's first **retail-accessible, P2P carbon credit exchange** enabling:

- Farmers / project owners (sellers)
- Corporates / individuals (buyers)

To **trade verified carbon credits (CCCs)** in a **transparent, real-time marketplace**.

Key Features & Functional Requirements

Onboarding & Security

- Phone Authentication: OTP-based login for mobile accessibility.
- Tiered KYC: Document upload (PAN/Aadhar) for regulatory compliance.
- Role-Based UI: Specific dashboards for Buyers and Sellers.

Real-time Carbon Exchange

- Live Order Book: Real-time visibility into Bids (Buyers) and Asks (Sellers).
- Price Index: A live tracking chart of the Carbon Credit Index in INR.
- Matching Engine: Algorithmic matching of buy/sell orders based on price and volume.

Wallet & Settlement

- Dual Asset Wallet: Management of both INR (Fiat) and CCCs (Carbon Credits).
- T+1 Settlement: Secure holding of funds during the verification window.
- Transaction History: Detailed logs for auditing and tax purposes.

Technical Architecture

- Frontend: React.js with Tailwind CSS (Optimized for Mobile/Capacitor).
 - Backend/Database: Firebase Firestore for real-time state synchronization.
 - Real-time Protocol: Firestore Subscriptions (onSnapshot) to replace traditional polling.
 - Deployment: Capacitor bridge for native iOS/Android binary generation.
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1.2 Objectives

- Enable **frictionless onboarding (KYC compliant)**
- Provide **real-time trading (like stock exchange UX)**
- Ensure **secure wallet + settlement system**
- Build **mobile-first scalable trading infra**

1.3 Target Users

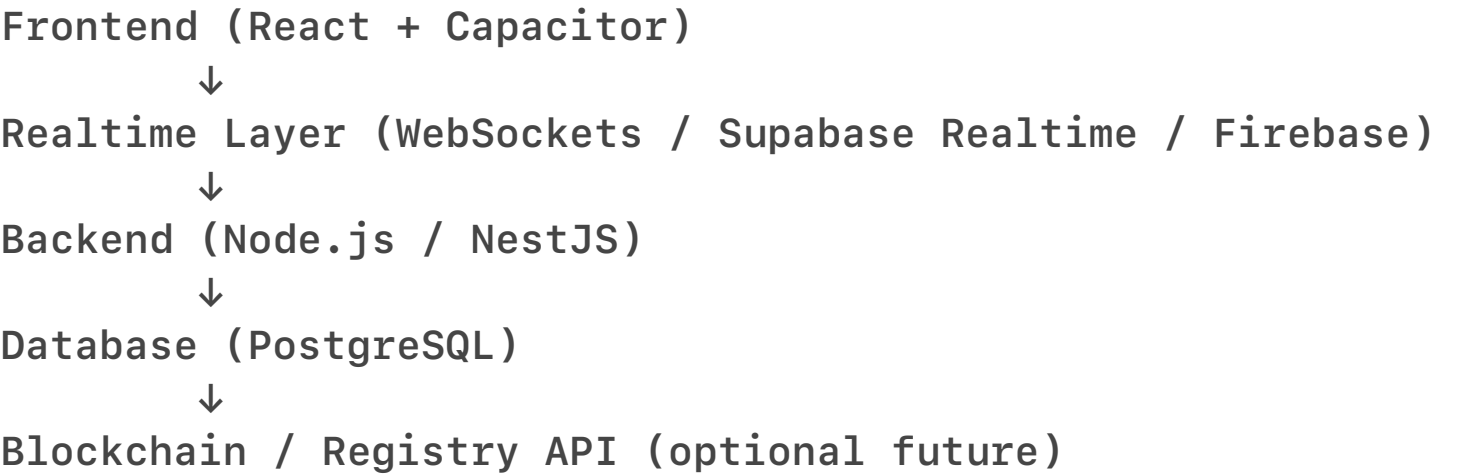
Persona	Description
Buyer	Corporates, ESG investors
Seller	Farmers, NGOs, renewable project owners
Admin	Platform regulators

2. 📱 Platform Scope

2.1 Platforms

- iOS (via Capacitor)
- Android (via Capacitor)
- Web (React – already built)

2.2 Architecture Overview



3. Core Features

3.1 Authentication & Onboarding

Features

- Phone number login (OTP)
- Role selection:
 - Buyer
 - Seller
- Session persistence (JWT)

Enhancements

- Firebase Auth / OTP Gateway
 - Biometric login (FaceID / Fingerprint via Capacitor)
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3.2 KYC Module

Steps

1. PAN Upload
2. Aadhaar Verification (API-based)
3. Bank Linking

Requirements

- AES-256 encryption
 - Integration with:
 - Digilocker
 - Aadhaar eKYC API
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3.3 Wallet System

Features

- INR Balance

- CCC Holdings
- Transaction History

Actions

- Deposit (UPI / NetBanking)
- Withdraw
- Settlement (T+1)

Backend Requirements

- Ledger-based accounting system
 - Double-entry bookkeeping
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3.4 Market (Core Trading Engine)

3.4.1 Order Book

Current Code

- Simulated order book (random updates)

PRD Requirement → REAL-TIME ORDER BOOK

Functional Requirements

- Live bids/asks
 - Depth visualization
 - Spread calculation
 - Matching engine
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3.4.2 Real-Time Sync (CRITICAL)

Implementation Options

- Supabase Realtime
 - Firebase Realtime Database
 - WebSockets ([Socket.io](https://socket.io/))
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Data Flow

Order placed →

Stored in DB →

Broadcast via Realtime subscription →

All clients update instantly

Example Schema

```
orders (  
  id,  
  user_id,  
  type (BUY/SELL),  
  price,  
  quantity,  
  status,  
  created_at  
)
```

Frontend Subscription (Example)

```
useEffect(() => {  
  const channel = supabase  
    .channel('orderbook')  
    .on('postgres_changes', { event: '*', schema: 'public',  
table: 'orders' }, payload => {  
      updateOrderBook(payload);  
    })  
    .subscribe();  
  
  return () => supabase.removeChannel(channel);  
}, []);
```

3.5 Price Engine

Current

- Static mock prices

Required

Real-Time Price Index Calculation

Last Traded Price (LTP)

VWAP (Volume Weighted Avg Price)

24h High/Low

Formula (VWAP)

$$\text{VWAP} = \frac{\sum (\text{Price} \times \text{Volume})}{\sum \text{Volume}}$$

Features

- Candle charts
- Intraday trend
- % change

3.6 Trade Execution Engine

Order Types

- Market Order
- Limit Order

Matching Logic

- Price-time priority

Execution Flow

Buy $\geq$ Sell price $\rightarrow$ Match

↓

Update wallets

↓

Record trade

↓

Notify users

3.7 Notifications System

Types

- Order executed
- Price alerts
- KYC updates

Tech

- Firebase Cloud Messaging (FCM)
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3.8 Profile & Settings

- KYC status
 - Bank details
 - Security settings
 - Logout
-

4. Native Mobile Conversion (Capacitor)

4.1 Setup Steps

```
npm install @capacitor/core @capacitor/cli
```

```
npx cap init
npx cap add ios
npx cap add android
npx cap sync
```

4.2 Native Features to Enable

Feature	Plugin
Camera (KYC upload)	Capacitor Camera
Biometric Auth	Capacitor Biometrics
Push Notifications	FCM Plugin
Storage	Secure Storage

4.3 Performance Optimization

- Lazy loading screens
- Reduce bundle size
- Use native navigation (optional upgrade)

5. 🗄 Database Design

Core Tables

Users

id, phone, role, kyc_status

Wallets

user_id, balance, credits

Orders

id, type, price, quantity, status

Trades

id, buyer_id, seller_id, price, qty

6. Security & Compliance

- AES-256 encryption
- HTTPS everywhere
- RBI/SEBI compliance (future)
- Fraud detection system

7. Analytics & KPIs

Metric	Goal
Daily trades	Growth
Liquidity	High
Avg trade size	Stable
Active users	Increase

8. Roadmap

Phase 1 (MVP)

- Auth + KYC
 - Wallet
 - Basic trading UI
 - Simulated order book
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Phase 2 (REAL PRODUCT)

- Realtime trading
 - Matching engine
 - Payment integration
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Phase 3 (ADVANCED)

- AI price prediction
 - Carbon project verification
 - Blockchain registry
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9. ⚠️ Risks & Challenges

- Regulatory uncertainty in India
 - Liquidity bootstrap problem
 - Fraudulent carbon credits
 - Real-time infra scaling
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10. 🎯 Success Criteria

- <100ms order latency
- Real-time sync across users
- Secure financial operations

- High retention (daily traders)

11. 🛠️ Suggested Tech Stack

Layer	Tech
Frontend	React + Tailwind
Mobile	Capacitor
Backend	Node.js / NestJS
Realtime	Supabase / WebSockets
DB	PostgreSQL
Payments	Razorpay

12. 💡 Key Improvements Over Current Code

Current

- ❌ Mock data
- ❌ No backend
- ❌ No persistence

After PRD

- ✅ Live trading system
- ✅ Real-time order book
- ✅ Native mobile apps
- ✅ Secure wallet

13. 🧠 Final Insight

This is **not just an app** — it's essentially a:

👉 **Mini stock exchange for carbon credits**

Which means your biggest focus should be:

- **Liquidity**
- **Trust**
- **Real-time performance**